

Total No. of Questions : 10]

SEAT No :

P2949

[5154]-501

[Total No. of Pages :3

B.E.(Civil)

ENVIRONMENTAL ENGINEERING-II
(2012 Course) (End Sem.) (401001) (Semester-I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8 and Q.9 or Q.10.*
- 2) *Figures to the right indicate full marks.*
- 3) *Draw neat figures wherever necessary.*
- 4) *Assume necessary data.*
- 5) *Use of scientific calculators is allowed.*

Q1) a) Differentiate between separate and combined sewerage system. **[5]**

b) Explain the procedure for determination of BOD. **[5]**

OR

Q2) a) What are the sources of sewage and how sewage flow quantity is estimated? **[5]**

b) Explain the procedure for determination of COD. **[5]**

Q3) a) Determine diameter and depth of primary sedimentation tank for sewage flow 10 million liters per day. **[5]**

Given data.

- i) Detention time = 2.5 hours
 - ii) Surface loading rate = 40000 l/m²/d
- b) Write streeter-Phelps equation, explain the terminology and write its application. **[5]**

OR

P.T.O.

Q4) a) An average operating data for conventional activated sludge treatment plant is as follows **[2+2+2]**

- i) Sewage flow = 30000 m³/d
- ii) Volume of aeration tank = 10500 m³
- iii) Influent BOD = 200 mg/l
- iv) Effluent BOD = 20 mg/l
- v) Mixed liquor suspended solids = 3000 mg/l
- vi) Effluent suspended solids = 30 mg/l
- vii) Waste sludge suspended solids = 9500 mg/l
- viii) Quantity of waste sludge = 200 m³/d

Determine.

- 1) Food to microorganisms ratio
 - 2) Sludge age
 - 3) Percentage of efficiency of BOD removal
- b) Differentiate between single stage and two stage filter. **[4]**

Q5) a) Design an oxidation pond for following data **[8]**

- i) Location : 24° Latitude
 - ii) BOD loading at 24° Latitude : 225 kg/ha/d
 - iii) Elevation : 900 m above sea level
 - iv) Mean monthly temperature : 30° maximum and 15° minimum
 - v) Population to be served : 10000
 - vi) Sewage flow : 100 lpcd
 - vii) Desired effluent BOD₅ : 20 mg/L
 - viii) Pond removal constant at 20°C : 0.1/d
- b) Write wastewater treatment principle of phytoremediation technology and explain its working with schematic sketch. **[4+4]**

OR

Q6) a) A wastewater flow is 10000 m³/d, BOD₅ is 200 mg/L, design an aerobic flow through type lagoon to serve a town of 50000 persons, using a ideal complete mixing model. Since the lagoon is proposed to be followed by another treatment unit, its size can be restricted to give a detention time of only 3 days. **[8]**

Given data

$k=0.015$ per day at 20°C, $Y = 0.5$, $k_d = 0.07$ per day

- b) Write wastewater treatment principle of root zone cleaning system and explain its working with schematic sketch. **[4+4]**

- Q7)** a) Draw a neat sketch of upflow sludge blanket (UASB) reactor. Explain the principle of working and its advantages and disadvantages. **[2+2+4]**
- b) The sludge is known to be 70% organic and 30% inorganic in nature. Approximately 60% of the organic fraction is converted to liquid and gaseous end products after a 30 day period. The digested sludge has a solids content of 5% and must be stored for periods of up to 85 days. Determine the volume of requirement for a standard rate single stage digester. The raw sludge loading rate is $80 \text{ m}^3/\text{d}$. **[8]**

OR

- Q8)** a) Write principle and stages of anaerobic digestion. Explain factors affecting digestion process. **[4+4]**
- b) Explain any two methods of sludge disposal with advantages, disadvantages and application. **[4+4]**
- Q9)** a) Explain methods of waste water sampling. **[6]**
- b) Write short note on equalization and neutralization. **[6]**
- c) Draw and explain units of treating dairy wastewater. **[6]**

OR

- Q10)** a) Explain the following points related to sugar industry. **[4+3+3]**
- i) Flow sheet of manufacturing process and wastewater generation
- ii) Characteristics of wastewater
- iii) Flow sheet of wastewater treatment
- b) Explain in brief primary and secondary treatment process adopted for treating industrial wastewater. **[4+4]**

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